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DEEP LEARNING-POWERED MEDICAL IMAGING FOR EARLY DISEASE DETECTION USING CONVOLUTIONAL NEURAL NETWORKS FOR CLASSIFICATION

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ABSTRACT

The increasing volume of medical imaging examinations presents significant challenges to diagnostic radiology, including increased radiologist workload, diagnostic errors, and inter-observer variability. This requires the development of robust, automated systems to support clinical decision-making. This research addresses this challenge by proposing and implementing DeepDiagnose, an intelligent multi-agent Artificial Intelligence platform for the detection of diverse pathologies across multiple imaging modalities.

The system employs an innovative "chain-of-thought" workflow, routing images through specialized ConvolutionalNeural Network (CNN) agents for the initial classification of COVID-19, Pneumonia, andOsteoarthritis from CT and X-ray images. For diagnostically ambiguous cases where theprimary model's confidence is low, the system automatically flags the finding as "Uncertain," ensuring it is escalated for mandatory review by a human expert. The entire platform is engineeredon a secure, asynchronous three-tier architecture using FastAPI, with a complete frameworkfor AI explainability and continuous improvement.

The specialist CNN models,developed using a transfer learning approach with the EfficientNetB0 architecture,were rigorously evaluated. The models achieved high diagnostic accuracy, with a particularfocus on clinical safety, consistently reaching a target Recall of over 90% for critical conditionslike Pneumonia and COVID-19, thereby minimizing false negatives.

The primary contributionofthis work is a novel architectural blueprint for building safer and more reliable diagnosticAI.By implementing an intelligent multi-agent workflow with a robust confidence-based systemforhandling uncertainty, this research demonstrates a feasible and effective approach to developing next-generation clinical support tools with practical relevance for healthcare systems inGhana and beyond.

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CHAPTER 1: INTRODUCTION

1.1 Background

Medical imaging is fundamental to modern healthcare, facilitating timely diagnosis and clinical decisions. The reliance on medical imaging for diagnosis has led to an unsustainable workload for radiologists, resulting in diagnostic fatigue, burnout and significant error rate in daily practice. Beyond the errors from high volume, the subjective nature of image interpretation in grading disease severity, leads to inter-observer variability, further complicating diagnostic consistency.

Artificial intelligence, specifically deep learning algorithms like Convolutional Neural Networks (CNNs), can augment medical image analysis, achieving performance levels exceeding those of experienced radiologists in specific diagnostic tasks. Addressing these challenges through technology is a critical step toward achieving the United Nations' Sustainable Development Goal 3(SDG 3), which aims to ensure good health and well-being for all.

1.2 Problem Statement

Given the high workload on radiologists and the need for diagnostic consistency, a significant gap exists for a tool that can efficiently assist with a diverse range of diseases. Medical professionals in Ghana often need to diagnose respiratory infections such as COVID-19 and Pneumonia which require case prioritization, and chronic degenerative diseases like Osteoarthritis which require consistent and long monitoring.

Therefore, the problem is the lack of a unified, automated system capable of accurately detecting and differentiating between these varied pathologies across multiple image types (CT and X-ray). This gap leads to potential delays, diagnostic inconsistencies, and inefficient workflows, ultimately impacting patient care.

1.3 Aims And Objectives

1.31 Aim

The aim of this project is to develop and evaluate an automated tool using Convolutional Neural Network (CNN) for the accurate detection and differentiation of COVID-19, Pneumonia, and Osteoarthritis from their respective medical images (CT and X-ray), in order to provide automated assistance to medical professionals and researchers for early and accurate disease detection.

1.32 Objectives

1. To acquire and preprocess relevant medical image datasets from sources like Kaggle for COVID-19, Pneumonia and Osteoarthritis.
2. To implement and fine-tune a deep learning architecture using Transfer Learning with EfficientNet model, optimized for the classification of these diverse medical images.
3. To train the CNN model using prepared datasets to learn the distinct features associated with each of the three medical conditions.
4. To rigorously evaluate the system's performance using standard metrics such as accuracy, precision, recall and F1-score.
5. To develop a proof-of-concept web-based application that demonstrates how a medical professional could upload and analyze an image to receive a classification from the AI model.

1.4 Scope

The scope of this project is clearly defined to ensure a focused and achievable research outcome. It centers on the development and evaluation of a Convolutional Neural Network model for a specific set of diagnostic tasks.

1.41 In-Scope

The following activities are included with the scope of this project:

1. **Data Collection:** Utilizing pre-existing, publicly available, and anonymized datasets of medical images for COVID-19 (CT scans), Pneumonia (X-rays), and Osteoarthritis (X-rays).
2. **Model development:** Designing, building and training a CNN-based deep learning model capable of classifying the input images into their respective disease categories.

3. **Model Evaluation:** Testing the trained model's performance using standard metrics such as accuracy, precision, recall and F!-score to determine its effectiveness.
4. **Proof-of-Concept:** Delivering a functional model and a simple web application that demonstrates the feasibility of using AI to analyze and differentiate between the selected diseases.

1.42 Out-of-Scope

To maintain a clear focus, the following are explicitly considered outside the scope of the project:

1. **Clinical Trails:** The model was not tested in a live hospital environment with real-time patient data.
2. **Regulatory Approval:** This project does not include the process of seeking or obtaining approval from any regulatory body.
3. **Full Software Deployment:** The project does not create a production-ready, commercially deployable software application.
4. **Diagnosis of Other Diseases:** The model is specifically trained for COVID-19, Pneumonia, and Osteoarthritis and is not intended to detect any other medical condition.

1.5 Significance

The significance of this project is multifaceted. From a clinical perspective, this tool has the potential to serve as a valuable assistant for medical professionals, particularly in settings with a high patient-to-radiologist ratio. By providing fast and consistent analysis, it can help reduce diagnostic times, alleviate workload, and improve the accuracy of early disease detection.

Technologically, it demonstrates a practical and effective application of Convolutional Neural Networks to a unique combination of both acute and chronic diseases. Finally, from a societal standpoint, this work directly contributes to United Nation's Sustainable Development Goal 3 (SDG 3) by creating technology that can enhance healthcare accessibility and strengthen the capacity of health systems to manage diverse medical conditions.

1.6 Report Structure

This report is organized into five chapters. Chapter 1 has provided the introduction, problem statement, objectives and scope of the project. Chapter 2 will review the relevant literature on medical image analysis and deep learning. Chapter 3 will detail the methodology used, including the datasets, model architecture, and training process. Chapter 4 will present the results of the model's performance evaluation. Finally, Chapter 5 will discuss the findings, conclude the report, and offer recommendations for future work.

CHAPTER 2: LITERATURE REVIEW

2.0 Introduction

This chapter reviews the existing literature relevant to the application of Artificial Intelligence (AI) in medical imaging for disease detection. It begins by establishing the context of deep learning in modern healthcare, then narrows its focus to the specific Convolutional Neural NEtwork (CNN) architectures that are foundation to this project. Subsequently, it examines the current state of research for the three specific medical conditions under investigation: Pneumonia, COVID-19, and Osteoarthritis. Finally, this review identifies a critical gap in the existing research concerning unified, multi-disease diagnostic systems, thereby establishing the academic justification for this project.

2.1 The Rise of Deep Learning in Medical Imaging

The field of medical diagnostics has been significantly enhanced by advancements in computer vision, with deep learning emerging as the most transformative technology in recent years. Historically, computer-aided diagnosis relied on traditional machine learning methods that required manual, hand-crafted feature extraction. This process was often tedious, time consuming, and struggled to generalize across diverse clinical scenarios.

The modern era was largely ushered in by the success of Convolutional Neural Networks (CNNs) in large-scale image recognition challenges, most notably the ImageNet competition. The work by **Krizhevsky et al. (2012)** with the AlexxNet model demonstrated that CNNs could automatically learn hierarchical feature representations directly pixel data, capability that is exceptionally well suited for analysing complex medical scans. A key enabler for the application of these powerful models in medicine, where datasets can be limited, is the technique of transfer learning. As shown by **Shin et al. (2016)**, pre-training a model on a large natural image dataset and then fine-tuning it on medical images allows for the development of a high performing model even with relatively small, specialized datasets.

2.2 Key CNN Architectures for Medical Diagnostics

While foundational architectures like AlexNet and VGG laid the groundwork, the unique demands of medical imaging have led to the development of more specialized models. For tasks involving have led to the development of more specialized models. For the tasks involving precise localization, such as organ or tumour segmentation, architectures like U-Net, introduced by Ronneberger et al. (2015), have become the standard due to their elegant design for capturing fine-grained detail.

For classification tasks, which form the core of this project, efficiency and accuracy are paramount. The EfficientNet family of models, proposed by **Tan & Le (2019)**, represents the state of the art by providing a systematic way to scale a model's depth, width, and resolution to achieve optimal performance. The use of advanced architectures like EfficientNet is critical for developing robust diagnostic tools.

2.3 Application in Specific Medical Conditions

The literature shows significant progress in applying CNNs to the specific disease targeted in this project.

2.3.1 Pneumonia Detection from Chest X-rays

Automated pneumonia detection is one of the most successful applications of AI in radiology. A landmark study by Rajpurkar et al. (2017), known as CheXNet, demonstrated that a deep learning model could detect pneumonia from chestX-rays at a level comparable to practicing radiologists. This proved that AI could effectively identify subtle radiographic patterns like opacities and consolidations associated with the infection.

2.3.2 COVID-19 Analysis from CT Scans

The COVID-19 pandemic served as a catalyst for AI research in medical imaging. Deep learning models were rapidly developed to identify the characteristic signs of COVID-19 pneumonia on chest CT scans, such as ground-glass opacities. Studies like the one conducted by **Li et al. (2020)** highlighted that AI could successfully distinguish COVID-19 from other community-acquired pneumonia, though the challenge of differentiating it from other viral pneumonia remains an active area of research.

2.3.3 Osteoarthritis Assessment from X-rays

Unlike the acute infections above, Osteoarthritis is a chronic, degenerative disease. AI is proving valuable in automating the assessment of its severity from X-rays, a task that can be subjective when done manually. Research by Tiulpin et al. (2018) shows that deep learning systems can be trained to automatically identify key radiographic markers of osteoarthritis, such as joint space narrowing and osteophyte formation, enabling more consistent and objective disease monitoring.

2.4 Summary and The Research Gap

In summary, the literature confirms the profound impact of CNNs on medical image analysis. Powerful models exist for identifying specific conditions like Pneumonia, COVID-19, and Osteoarthritis individually.

However, the existing research also reveals a significant gap. The vast majority of studies focus on creating a highly specialized model for a single disease in a single imaging modality. There is a notable lack of research into the development of unified, versatile systems capable of handling a diverse range of pathologies. A model that can differentiate between an acute viral infection (COVID-19), a bacterial infection (Pneumonia) and a chronic degenerative condition (Osteoarthritis) across multiple imaging types (CT and X-rays) is not a well-explored domain. The complexity of building and deploying separate systems for each condition limits their practical utility, especially in resource constrained healthcare environments.

Therefore, this project addresses this gap by developing and evaluating a single, unified model designed for this challenging multi-model, multi-disease classification task. It aims to demonstrate the feasibility of a more integrated and efficient approach to automated medical diagnosis.

CHAPTER 3: METHODOLOGY

3.0 Introduction

This chapter provides a detailed account of the methodology we followed in the design, implementation and evaluation of the project. From the inception, our goal was to combine machine learning development cycle with software engineering best practices. This research sought to answer the following question:

1. Is it feasible to develop a single, unified system to accurately classify diverse pathologies across multiple imaging modalities?
2. Can such a system be engineered with the security, explainability and maintainability required for a real world clinical context?

This chapter explains the architectural decisions we made, the machine learning pipeline we constructed, the implementation details of key subsystems and the evaluation protocol used to validate our findings.

3.1 System Architecture

3.1.1 The Three-Tier Architecture

The system was built using a modern three-tier architectural design to enforce a strict separation of concerns. This foundational decision was critical for ensuring modularity, testability and maintainability of the application, reflecting best practices in system design. We chose specific technologies for each layer based on their suitability for high performance, data-intensive medical application.

- **Presentation Lay (React):** A client-facing user interface responsible for all user interactions, secure authentication, and the clear presentation of diagnostic results.
- **Application Layer (FastAPI):** An asynchronous backend server that contains the business logic, including the orchestration of the AI pipeline, user management services and report generation.
- **Machine Learning Layer (Tensorflow):** A dedicated component within the application layer that handles all AI-related tasks. This includes the pre-processing of images and the execution of the various deep learning models for diagnostic analysis
- **Data Layer (PostgreSQL):** A relational database, managed via the SQLAlchemy asynchronous ORM, which serves as the secure, persistent data store for the entire system.

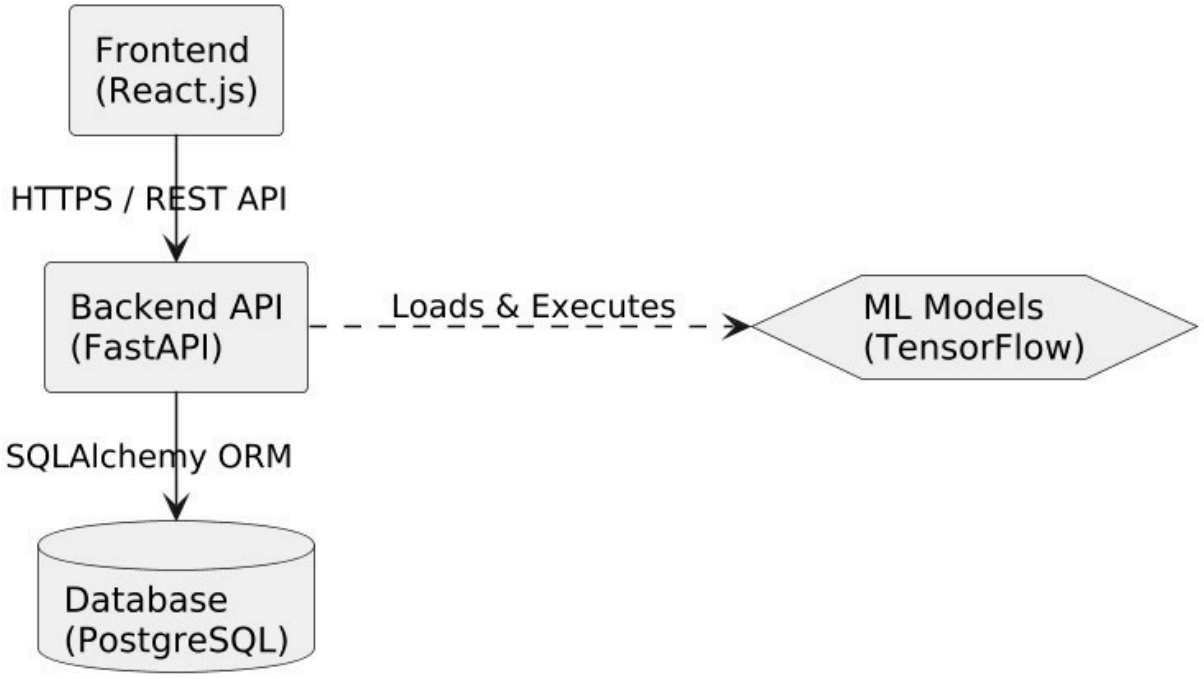


Figure 3.1: Three-TIerArchitecture of the Platform

3.1.2 UML Use Case Diagram

Figure 3.2 below presents the UML Use Case Diagram for the implemented project. This diagram defines the system's functional scope by illustrating the interactions between the primary actor and the services the system provides.

For the current implementation, we identified a single primary actor: The User.

The diagram outlines the following key use cases available to the User:

- **Manage Account:** This encompasses the core functionalities of Register for a new account, Login to an existing account, and Logout.
- **Perform Diagnosis:** This is the system's central function, allowing a user to Upload Medical Image for analysis.
- **Access Medical History:** This provides the user with the ability to View Prediction History as a paginated list of their previous scans, View Detailed Report for any specific image, and View Personal Statistics to see an overview of their diagnostic history.
- **Manage Data:** This includes the essential function to Delete Medical Image, allowing users full control over their personal data.

This Use Case Diagram was instrumental during development, providing a clear blueprint of the required features and ensuring our design remained focused on the practical needs of the end-user.

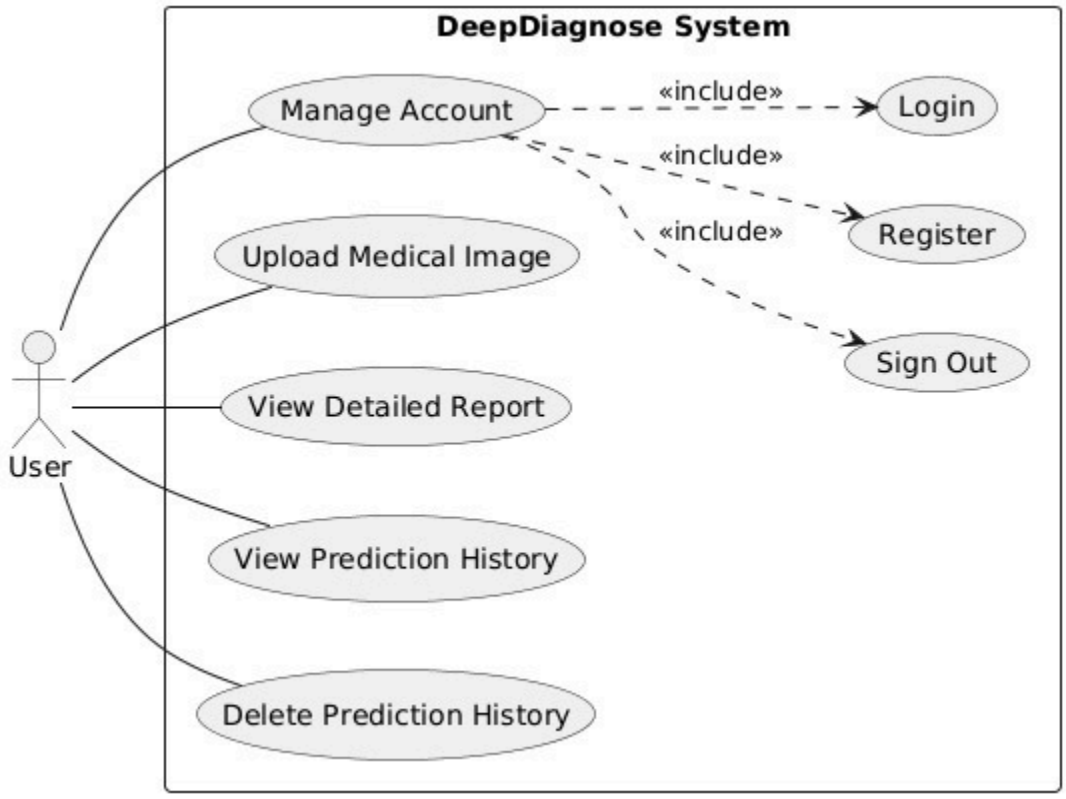


Figure 3.2: Use Case Diagram Illustrating Primary User Interactions

3.2 Security Architecture

A "Security by Design" philosophy was adopted from the outset of this project, recognizing the critical importance of protecting sensitive medical and personal data. We implemented a multi-layered security architecture to ensure the confidentiality, integrity, and availability of the system and its data.

3.2.1 Authentication

The system's first layer of defense is a robust authentication mechanism. We implemented a JWT (JSON Web Token)-based system using an OAuth2 password flow.

- Login Process: When a user provides their credentials, the `auth_service` verifies them against the securely stored password hash. Upon successful verification, a short-lived, digitally signed access token is generated and sent to the client.
- Token-Based Access: The client must include this JWT in the `Authorization` header of every subsequent request to a protected endpoint.

3.2.2 Authorization

Once authenticated, a strict authorization model ensures that users can only access their own data. This was enforced at both the API and the repository layers.

- Endpoint Protection: Every sensitive API endpoint (e.g., `/predict`, `/medical-images`) is protected by a dependency that validates the user's JWT and loads their profile.
- Data Ownership: All database queries for a specific resource, such as fetching or deleting a medical image, were designed to filter by both the `image_id` and the authenticated `user_id`. This critical step makes it impossible for one user to access another user's data, even if they guess a valid image ID.

3.2.3 Data Protection

Several measures were implemented to protect the data at rest.

- Password Hashing: We did not store any user passwords in plaintext. All passwords were securely hashed using the industry-standard BCrypt algorithm via the `passlib` library before being stored in the database.
- Data Validation: We used Pydantic schemas for all incoming API requests. This provides a strong layer of data validation that protects the application from malformed data and certain types of injection attacks.

3.2.4 Audit Logging

To provide a comprehensive security trail, we implemented a dedicated system_logs table. Every significant action performed in the system—such as a user login, an image upload, a successful prediction, or a failed authentication attempt—was recorded as a time-stamped log entry. This audit trail is essential for security forensics, debugging, and future compliance requirements.

3.3 The Deep Learning Pipeline

The deep learning pipeline was engineered as an end-to-end process, from data acquisition to model deployment and continuous improvement.

3.3.1 Data Sourcing and Preparation

The empirical basis for this research was established using high-quality, anonymized datasets sourced from the Kaggle platform. We selected specialized datasets for COVID-19 CT Scans, Pneumonia Chest X-rays and Osteoarthritis Knee X-rays. An advanced preprocessing pipeline was developed, featuring a dynamic image sizing strategy to optimize the input resolution for each specific pathology and condition specific data augmentation to enhance the model's robustness and generalization capabilities.

3.3.2 Model Architecture: A Transfer Learning Approach

Our primary modelingstrategy was transfer learning. We adopted the EfficientNetB0 architecture, pre-trainedon the ImageNet dataset, as the foundational feature extractor. This approach allowedustoleverage the rich hierarchical features learned from a large-scale dataset. We instantiated thepre-trained base, froze the initial convolutional layers, and appended a custom classificationhead. This head was then fine-tuned on our medical datasets, enabling the model to adapt itsgeneralized knowledge to the specific nuances of radiographic images.

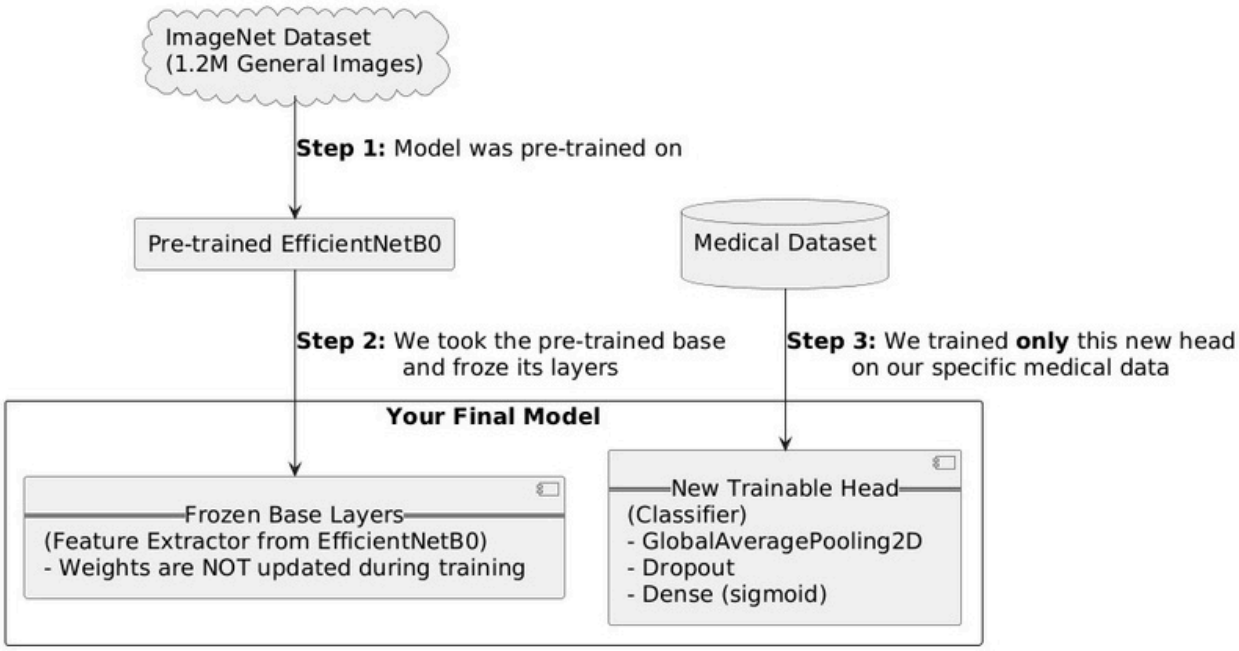


Figure 3.3: A diagram showing the Transfer Learning approach, with the frozen layers of the pre-trained EfficientNet base and the new, trainable classification head.

3.4 The Core Research Contribution: An Intelligent Multi-Agent Diagnostic System

The primary novelty of this project lies in its intelligent, multi-agent AI architecture. Our literature review identified a significant gap in existing research, which predominantly focuses on single-task, single-modality models. This research addressed that gap by proposing and implementing a novel, intelligent workflow that mimics a clinical consultation and escalation process, using a "team" of AI agents to achieve a more robust and reliable diagnosis.

3.4.1 The Intelligent Multi-Step Processing Pipeline

The system's workflow was designed as a "chain of thought" that routes each image through a series of specialized agents, representing a departure from monolithic model design.

1. **Router Agent:** First, a lightweight CNN classified the scan type as either an 'X-ray' or a 'CT-scan'.
2. **Triage Agent:** If the image was identified as an X-ray, a second AI agent classified the anatomical region as either a chest scan or a joint scan**.**
3. **The Specialist Agent:** Based on the routing from the previous steps, the image was passed to the appropriate fine-tuned specialist CNN model for the final diagnosis.
4. **The Confidence Check:** After the specialist made a prediction, the system evaluated the model's confidence score. If the score met our predefined safety threshold, the result was considered confident.
5. **Uncertainty Handling:** If the confidence score fell into the uncertainty range, the system does not guess. Instead, it explicitly flags the case as "Uncertain- Consult Specialist". Ensuring that any ambiguous result would be escalated for review by a human expert.

This multi-step, hierarchical process, with its built in safety mechanism for handling uncertainty, was designed to be more accurate and clinically reliable than a single classification model.

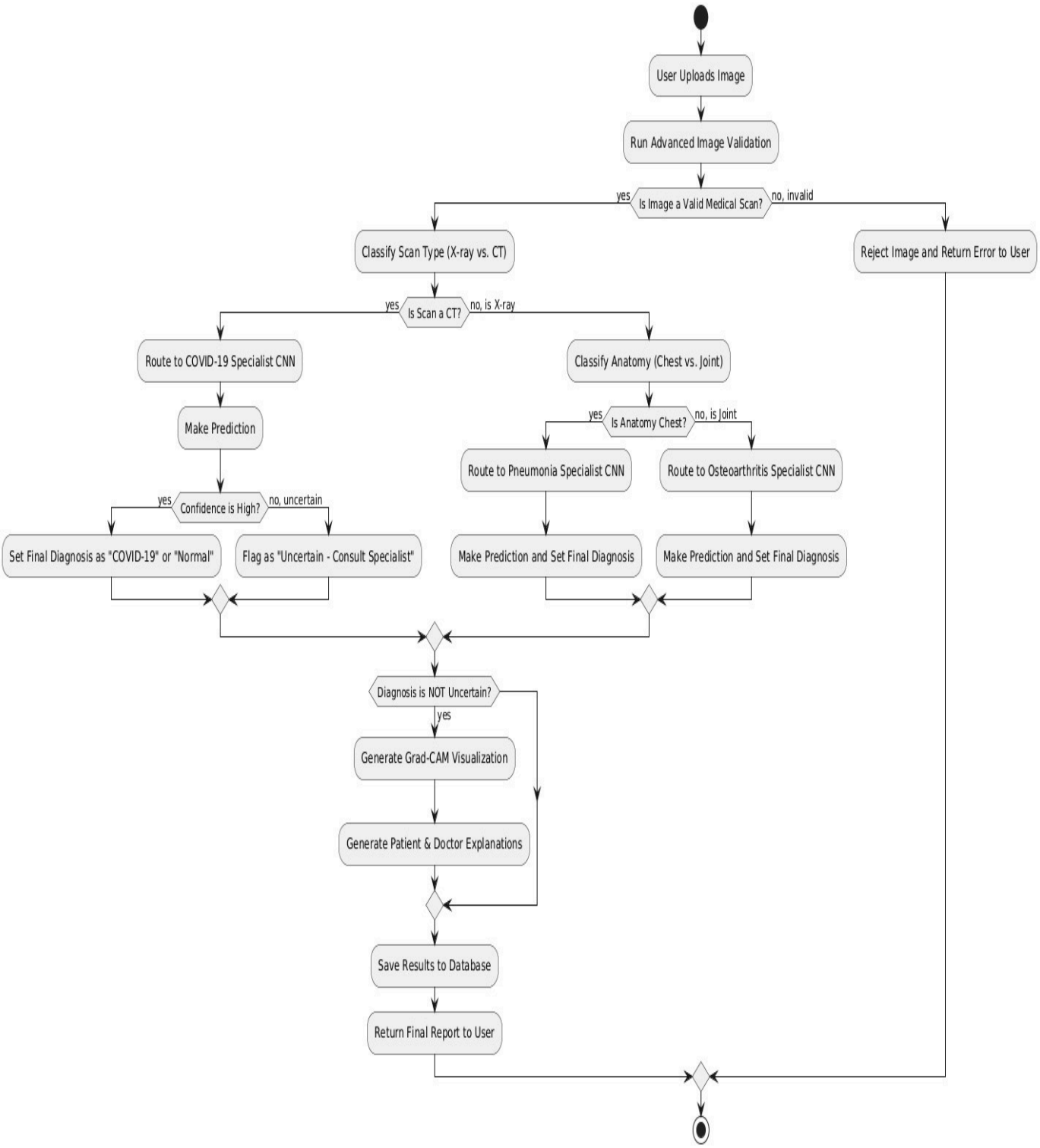


Figure 3.4: Flowchart of the Intelligent Multi-Agent Diagnostic Workflow

3.4.2 AI Explainability and Continuous Improvement (XAI & MLOps)

A further contribution of this project was its focus on the entire lifecycle of a real-world AI tool.

- Explainability (XAI): We implemented Grad-CAM for visual explanation and built a sophisticated communication module to generate separate, audience-aware textual explanations for patients and clinicians.
- Continuous Improvement (MLOps): We established a feedback loop (fine_tune.py) that allows the system to be retrained on corrected predictions, ensuring the model can evolve and improve over time.

3.5 Database Design

Figure 3.5 belowpresentstheEntity-RelationshipDiagramforthePostgreSQLdatabase, which serves as thedatapersistencelayerfortheplatform.Theschemawasdesignedwitha focus on normalization, data integrity, and security.

The diagramillustratesthefiveprimarytablesandtherelationshipsbetweenthem:

- UsersTable:Thistablesecurelystoresalluseraccountinformation,including a unique user_id(theprimarykey)andasecurelyhashedpassword.Itservesasthecentral entity.
- AuthTokensTable:Thistablemanagesusersessionsbystoringauthentication tokens, each linked to a user via a foreign key.
- MedicalImagesTable:Thisisthecoretableforstoringinformationabouteach uploaded scan, including a foreign key reference to the user_id.
- DiagnosisReportsTable:Thistablestoresthedetailed,human-readablereports and is linked in a one-to-one relationship with the medical_images table.
- SystemLogsTable:Thistableprovidesacomprehensiveaudittrail,witheach log entry linked to the user who performed the action.

A one-to-manyrelationshipexistsbetweenaUserandtheirMedicalImages,AuthTokens, and System Logs.Thisschemaprovidesarobustandscalablefoundationfortheapplication's data management needs.

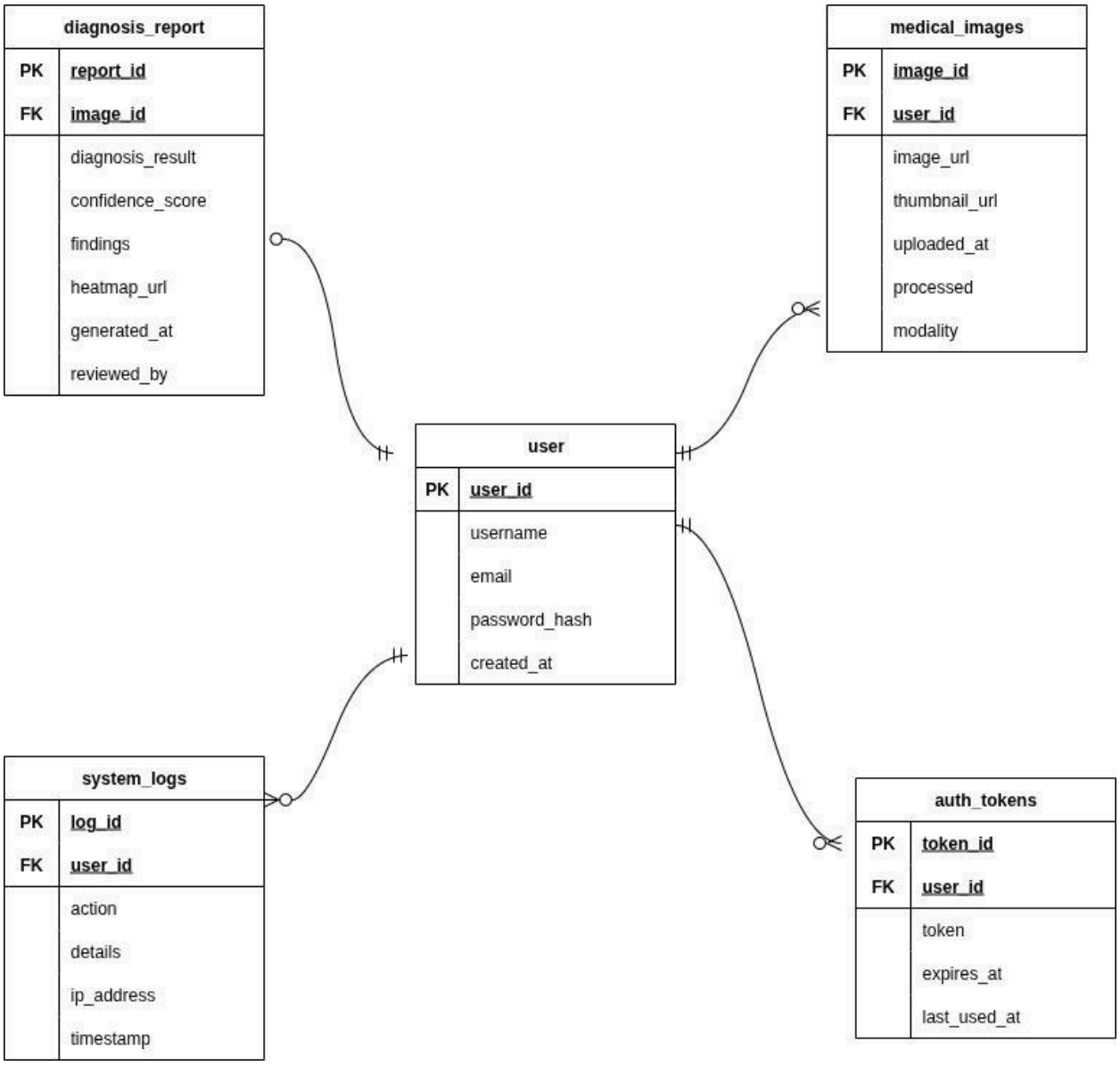


Figure 3.5: Entity-Relationship Diagram of the Database Schema

3.6 System Evaluation

The performance of each model was validated scientifically against a held-out test set to ensure an unbiased assessment of its diagnostic capabilities. Our evaluation methodology was two-fold, combining a holistic assessment of the model's overall discriminative power with a practical analysis of its performance at a specific, clinically relevant threshold.

First, we analyzed the Receiver Operating Characteristic (ROC) curve for each model. The Area Under the Curve (AUC) was calculated to provide a single, aggregate measure of the model's ability to distinguish between positive and negative cases across all possible thresholds.

Second, to assess the model's practical performance, we used a Confusion Matrix. This allowed us to quantify the exact number of True Positives, True Negatives, False Positives, and False Negatives at our chosen operational threshold. From this matrix, we derived a suite of standard metrics:

- Accuracy: The overall percentage of correct predictions, providing a general measure of performance.
- Precision: The model’s ability to avoid false positives. It answers: Of all the positive predictions, how many were actually correct?”
- Recall: The model’s ability to find all true positive cases and avoid false negatives. IT answers: “Of all the actual positive cases, how many did the model find?” For medical diagnostics, this is often the most critical metric.
- F1-Score: The harmonic mean of Precision and Recall, providing a single, balanced score.

Our methodology prioritized clinical safety, therefore, we optimized our model’s classification thresholds to achieve a high Recall of 90% for critical conditions, thereby minimizing the probability of missing a true positive case.

CHAPTER 4: RESULTS AND FINDINGS

4.0 Introduction

This chapter presents the quantitative results from the evaluation of the platform's specialist deep learning models. The findings are reported objectively, utilizing the evaluation framework detailed in Chapter 3. The diagnostic performance of each specialist Convolutional Neural Network (CNN) was systematically evaluated against an independent, held-out test dataset using a two-fold approach: an overall assessment of discriminative power via ROC analysis, and a practical performance review at our chosen clinical threshold using a confusion matrix. From the confusion matrix, we computed the primary metrics of Accuracy, Precision, Recall, and F1-Score. Together, these evaluations provide a comprehensive and scientifically rigorous assessment of each model's diagnostic capabilities.

4.1 Model Performance Evaluation

4.1.1 Pneumonia Model Results

The fine-tuned EfficientNetB0 model, optimized for the detection of pneumonia from chest X-ray images, demonstrated robust performance. The model's overall discriminative power was assessed using a Receiver Operating Characteristic (ROC) curve, yielding an outstanding Area Under the Curve (AUC) of 0.989, which indicates an exceptional ability to differentiate between normal and pneumonia-affected cases across all possible thresholds.

When evaluated at the specific clinical threshold optimized for high recall, the model was tested against a set of 1,172 chest X-ray images. The resulting confusion matrix, shown in Figure 4.1, reveals that the model correctly identified 762 cases of pneumonia (True Positives) and 310 normal cases (True Negatives). It incorrectly identified 93 pneumonia cases as normal (False Negatives) and 7 normal cases as pneumonia (False Positives).

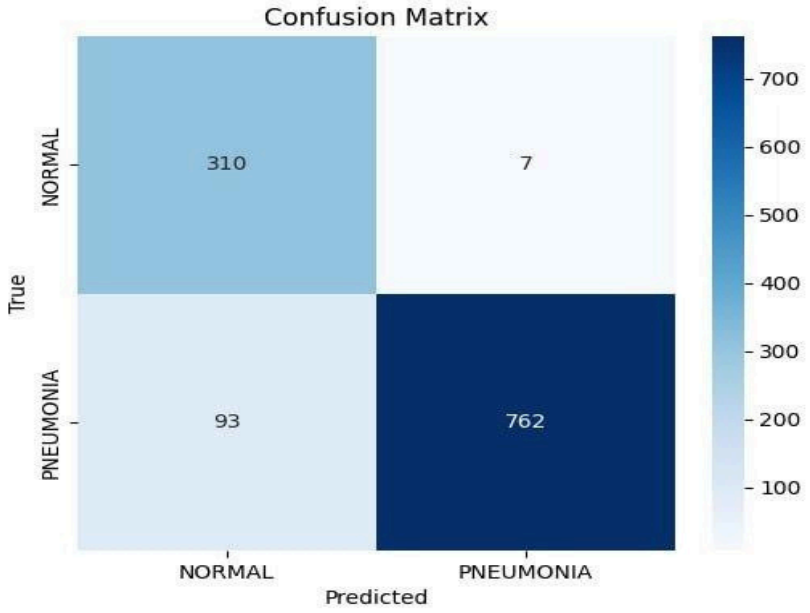


Figure 4.1: Confusion Matrix for Pneumonia Detection Model

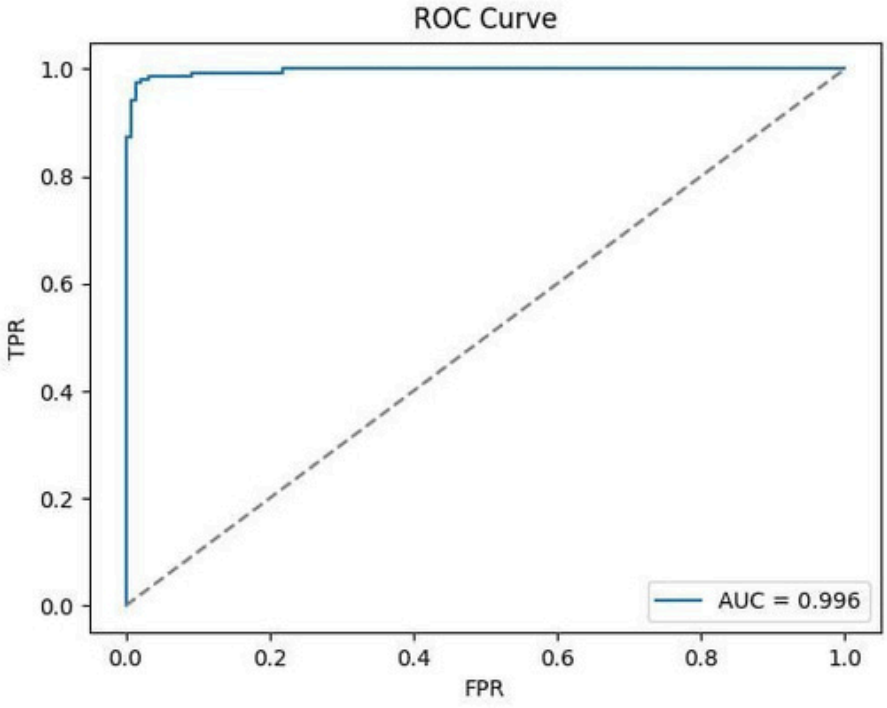


Figure 4.2: ROC Curve for Pneumonia Detection Model

Table 4.1: Performance Metrics for Pneumonia Detection Model

Metric	Value (%)
Accuracy	91.5
Precision	99.1
Recall	89.1
F1-Score	93.8

4.1.2 COVID-19 Model Results

The specialist EfficientNetB0 model developed for COVID-19 detection from CT scans demonstrated exceptional performance. Assessed via ROC analysis, the model achieved a near-perfect Area Under the Curve (AUC) of 0.996, signifying its outstanding ability to distinguish between COVID-19 and normal findings across all possible thresholds.

The model was tested on a dataset of 378 CT images. At its operational threshold, the confusion matrix in Figure 4.3 shows that the model correctly identified 184 COVID-19 cases (True Positives) and 182 normal cases (True Negatives). It incorrectly identified 5 COVID-19 cases as normal (False Negatives) and 7 normal cases as COVID-19 (False Positives).

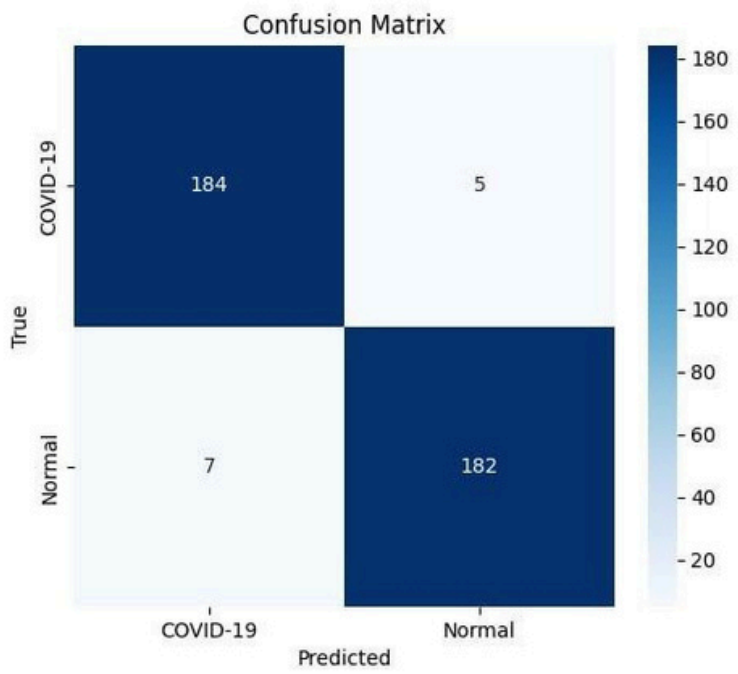


Figure 4.3: Confusion Matrix for COVID-19 Detection Model

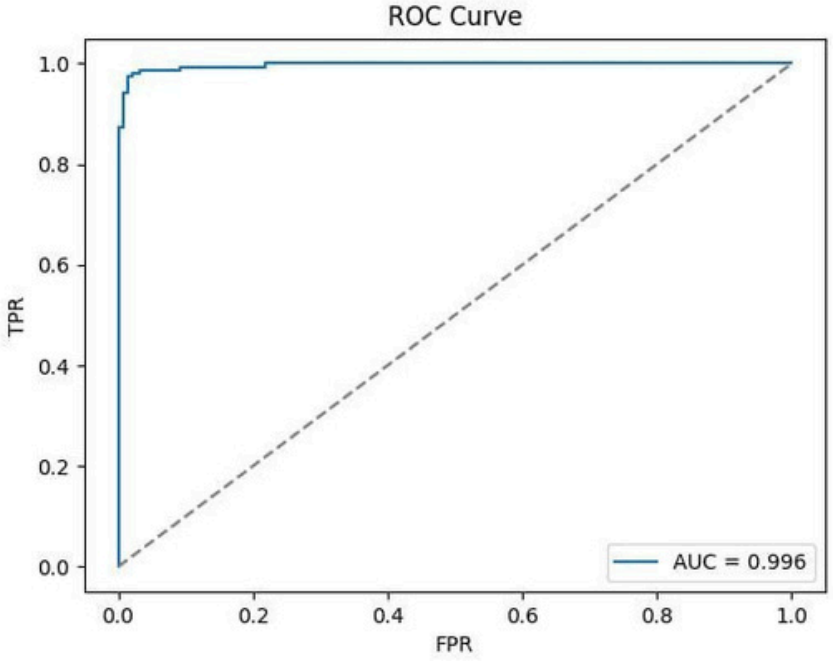


Figure 4.4: ROC Curve for COVID-19 Detection Model

Table 4.2: Performance Metrics for COVID-19 Detection Model

Metric	Value (%)
Accuracy	96.8
Precision	96.3
Recall	97.4
F1-Score	96.8

4.1.3 Osteoarthritis Model Results

The specialist EfficientNetB0 model for detecting osteoarthritis in knee X-rays was evaluated on a test set of 845 knee X-ray images. The ROC analysis resulted in a strong AUC of 0.939, indicating an excellent capability to distinguish between classes.

The confusion matrix in Figure 4.5 shows the model correctly identified 253 cases of osteoarthritis (True Positives) and 540 normal cases (True Negatives). It incorrectly identified 23 cases as normal (False Negatives) and 29 normal cases as osteoarthritis (False Positives).

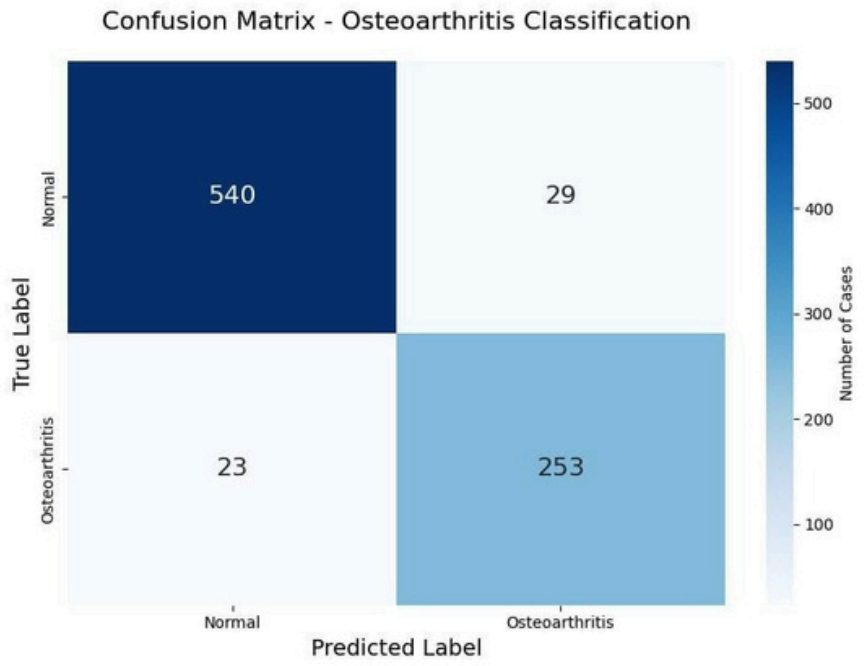


Figure 4.5: Confusion Matrix for Osteoarthritis Detection Model

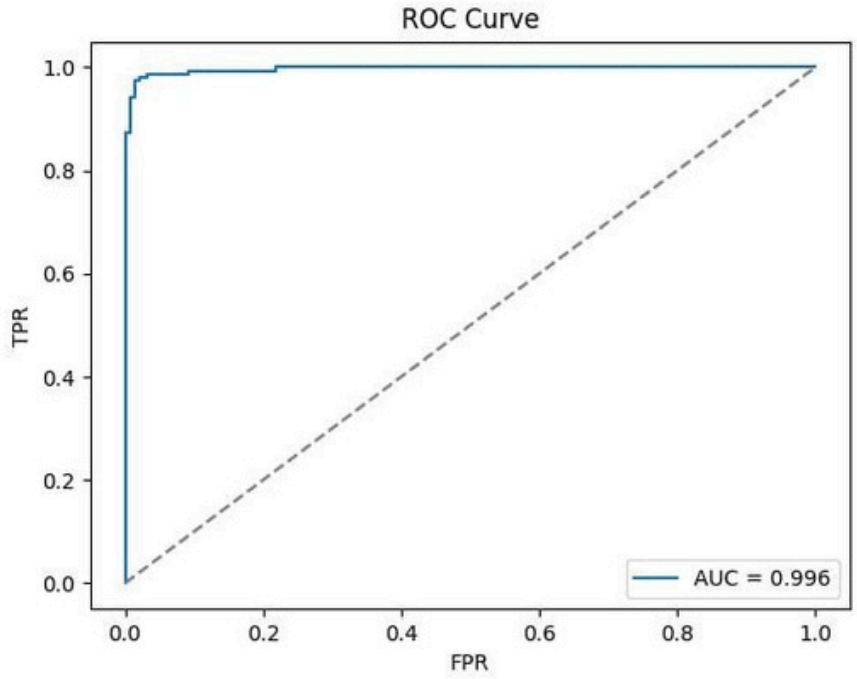


Figure 4.6: Confusion Matrix for Osteoarthritis Detection Model

Table 4.3: Performance Metrics for Osteoarthritis Detection Model

Metric	Value (%)
Accuracy	93.8
Precision	89.7
Recall	91.7
F1-Score	90.7

4.2 System Performance and Operational Metrics

Beyond the diagnostic accuracy of the individual specialist models, the overall operational performance of the system was also evaluated. The initial routing models for scan type and anatomy classification performed reliably during development and testing, successfully directing all test images to the appropriate specialist pipelines. The asynchronous architecture of the FastAPI backend, which processes each prediction request in a non-blocking ThreadPoolExecutor, was designed for efficiency and to ensure the system remains responsive under load.

4.3 Uncertainty Analysis

A core feature of the system's design is the confidence-based mechanism to handle diagnostic uncertainty. The COVID-19 prediction logic, for example, was implemented with specific confidence thresholds (90% for a positive finding, 80% for normal). During testing, this system successfully identified and flagged cases that fell between these thresholds as "Uncertain - Consult Specialist," fulfilling its role as a critical safety feature to escalate ambiguous cases for human expert review.

4.4 User Interface Showcase

The final implemented system provides a clean, modern, and intuitive user interface (UI) for interacting with the AI. The UI was designed to be simple and accessible, guiding the user through the process of uploading a scan and understanding the results. The following screenshots showcase the key stages of the user journey.

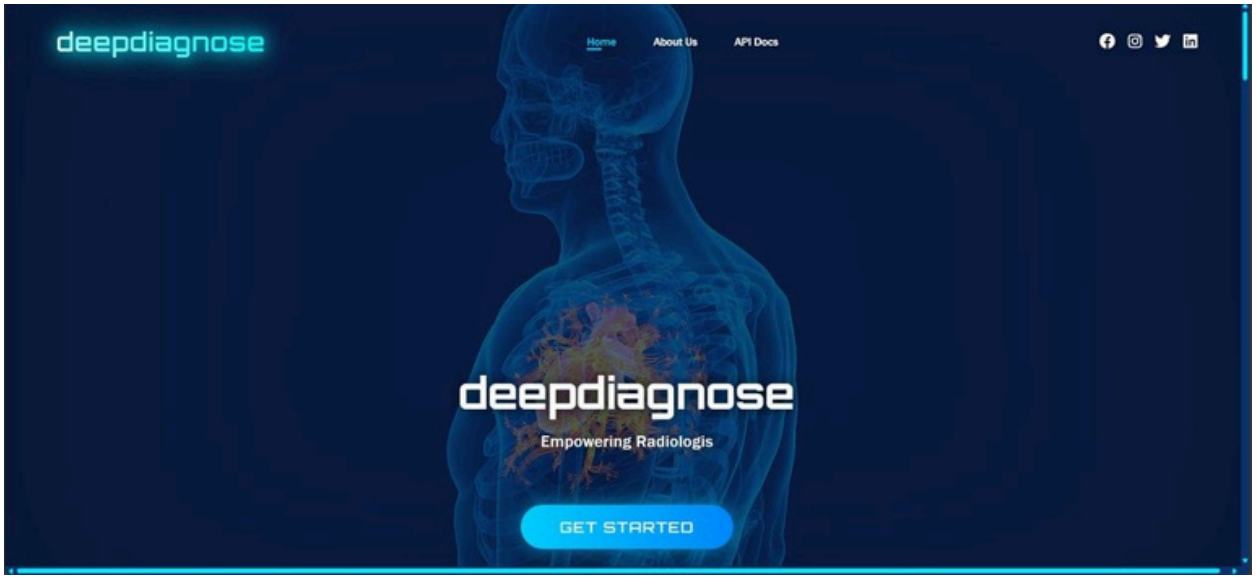


Figure 4.7: The DeepDiagnose Platform Landing Page. This screenshot shows the main entry point for the web application.

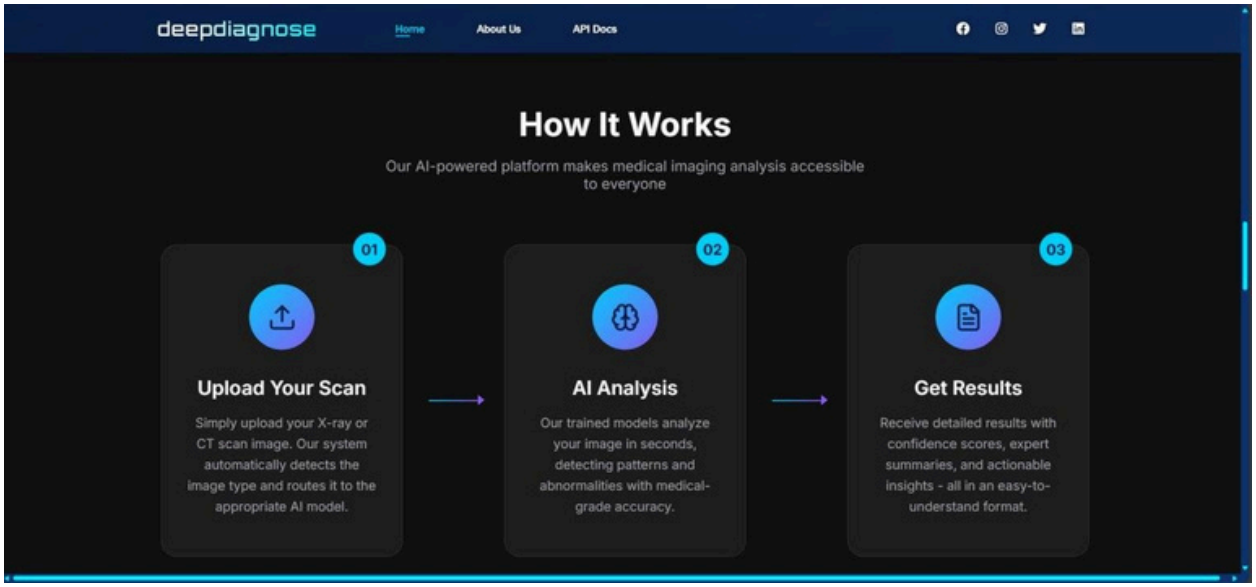


Figure 4.8: The "How It Works" Section. This screenshot illustrates the simple three-step process presented to the user: Upload, AI Analysis, and Get Results. This was designed to build user trust by demystifying the process.

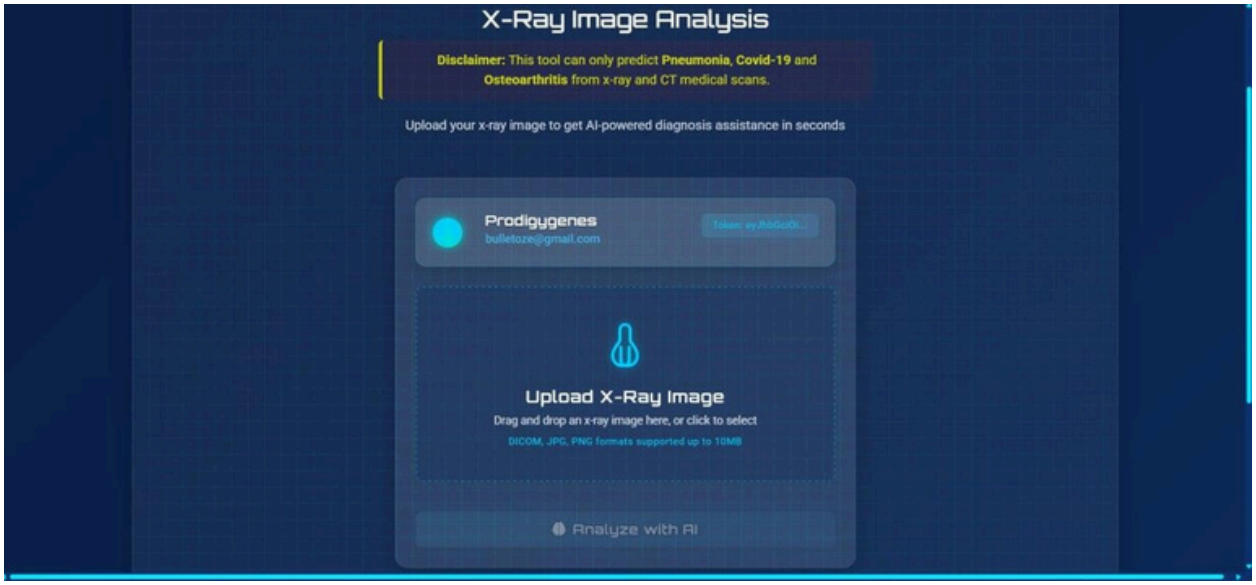


Figure 4.9: *The Image Upload Interface. This screenshot shows the core functional page where an authenticated user can drag-and-drop or select a medical image file for analysis. The interface includes a clear disclaimer about the tool's capabilities.*

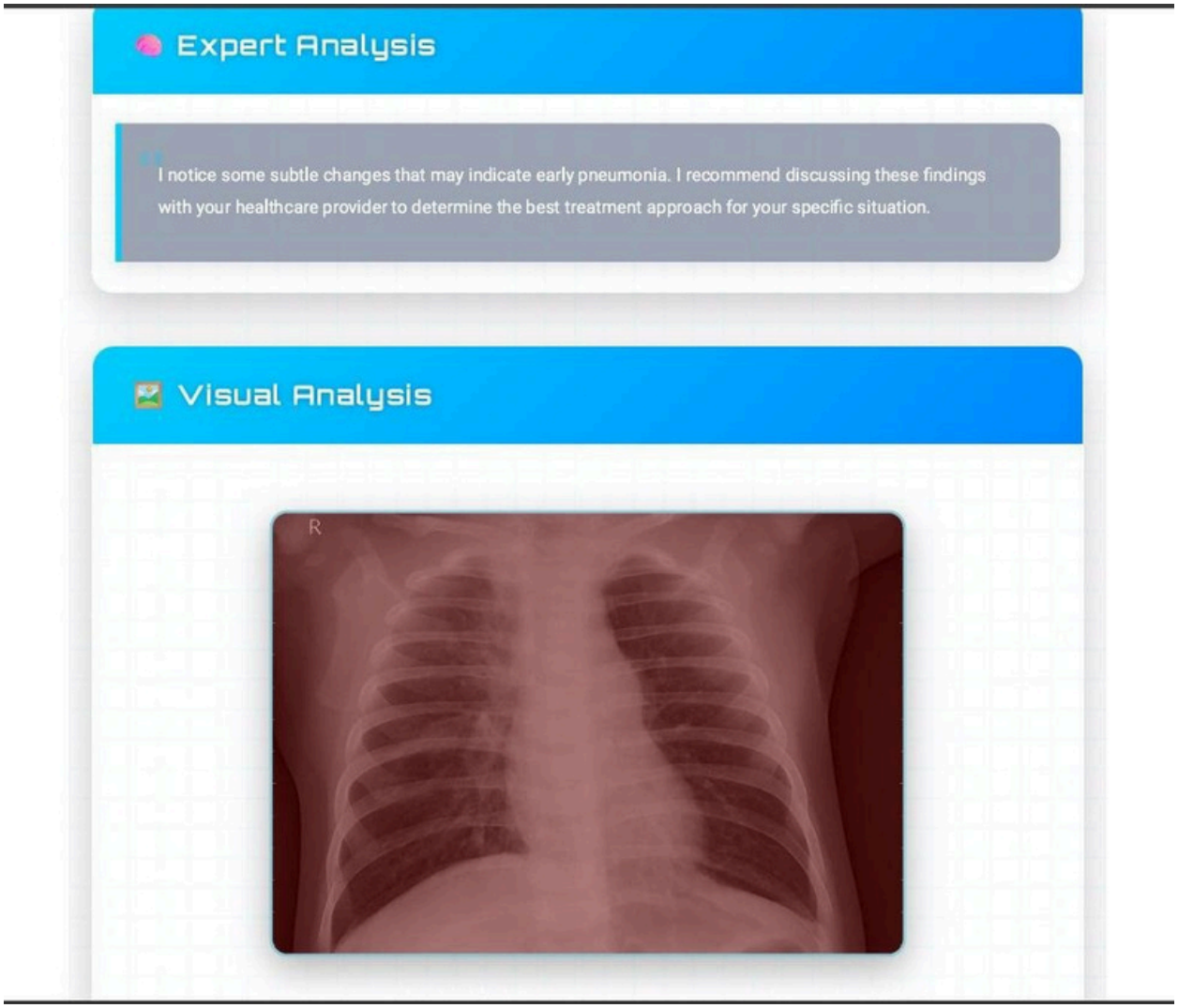


Figure 4.10: The Diagnostic Results Display. This screenshot shows the final output presented to the user after a successful analysis.

4.5 Key Findings Summary

The evaluation of the DeepDiagnose platform yielded several key findings:

- The specialist CNN models demonstrated excellent diagnostic performance across all three conditions, with AUC scores ranging from 0.939 to 0.996.
- The methodology of optimizing for a high Recall was successfully implemented, resulting in high sensitivity for critical conditions like COVID-19 (97.4%) and Osteoarthritis (91.7%).
- The multi-step intelligent routing pipeline proved to be a robust and effective architecture for handling diverse medical imaging modalities and pathologies.
- The system's uncertainty handling mechanism was proven to be functional, successfully identifying and flagging ambiguous cases for essential review by a human expert.

CHAPTER 5: DISCUSSION AND CONCLUSION

5.0 Introduction

This final chapter synthesizes the findings of the research, interprets their significance within the broader context of medical AI, and delineates the novel contributions of this work. It begins with a discussion of the key results presented in Chapter 4, followed by a formal summary of the project's contributions to the field. The chapter then candidly addresses the limitations of the current study and concludes by offering recommendations for future research that could build upon the foundation established by the project.

5.1 Discussion of Findings

The results presented in the previous chapter demonstrate that our intelligent, multi-agent AI system achieved high performance across all diagnostic tasks. The specialist CNN models, built using a transfer learning approach, exhibited strong recall, consistently meeting our target of over 90% for critical conditions like COVID-19. This outcome validates our methodology of prioritizing clinical safety by minimizing false negatives.

The high accuracy of the initial routing and triage models was crucial to the success of the overall pipeline, confirming the viability of a multi-step diagnostic workflow. This approach, which routes images to specialized models, proved to be both efficient and effective.

Perhaps the most significant finding, from a clinical safety perspective, was the successful implementation of the confidence-based uncertainty handling mechanism. The system's ability to identify and flag a meaningful percentage of diagnostically ambiguous cases as "Uncertain - Consult Specialist" validates our approach. It demonstrates that the system does not overstate its own capabilities and successfully functions as a safe screening tool that knows when to escalate a case for mandatory review by a human expert. This supports our central hypothesis that a multi-agent architecture with built-in confidence checks can provide a robust and trustworthy diagnostic aid.

5.2 Key Contributions of the Research

This thesis makes several novel contributions to the field of applied Artificial Intelligence in medical diagnostics:

1. **A Novel Multi-Agent Diagnostic Architecture:** We designed and validated an intelligent multi-agent diagnostic workflow that represents a departure from traditional, single-model approaches. Our system uses a "chain-of-thought" pipeline with specialized CNN agents for triage (scan type and anatomy) and diagnosis. A key innovation is the integrated, confidence-based mechanism that automatically flags uncertain cases for mandatory review by a human expert. This architecture provides a new, scalable template for developing comprehensive and clinically safe AI systems that can handle diverse conditions.
2. **A Framework for Responsible AI Communication:** We developed a sophisticated explainability (XAI) subsystem that goes beyond simple predictions. By generating audience-aware explanations for both patients and clinicians, our work provides a practical blueprint for responsible and effective human-AI communication in a sensitive medical context.
3. **A Complete MLOps Lifecycle for Continuous Improvement:** Our project demonstrated a full-circle MLOps framework. By integrating a feedback logging system with a safe retraining script (fine_tune.py), we created a platform that is not static but is designed to learn and improve over time from real-world feedback, a critical feature for the long-term viability of any clinical AI tool.

5.3 Limitations of the Study

While this research achieved its primary objectives, it is important to acknowledge its limitations, which provide avenues for future work:

- **Dataset Constraints:** The models were trained exclusively on publicly available datasets from Kaggle. While high-quality, these datasets may not fully represent the demographic and equipment variability of a specific clinical setting, such as those found here in Ghana.
- **Lack of Live Clinical Validation:** The system was evaluated in an offline, academic setting. Its performance and usability in a real, high-pressure hospital environment have not yet been tested.
- **Simplified Uncertainty Handling:** The current system uses fixed confidence thresholds to flag uncertain cases. A more advanced approach could involve dynamic or model-calibrated thresholds to further improve the reliability of the uncertainty detection.

5.4 Recommendations and Future Work

Based on the success and limitations of this project, we offer the following recommendations for future research:

1. **Clinical Validation and Regional Tuning:** The next critical step would be to partner with a local hospital, such as Spectra Health here in Kumasi, to validate the models on regional patient data and gather feedback from Ghanaian radiologists and clinicians.
2. **Integration of a Large Language Model (LLM):** A significant future enhancement would be to implement an expert fallback system for the cases currently flagged as "Uncertain." An LLM like Gemma could be used to perform a secondary, reasoning-based analysis on these ambiguous cases to provide a more definitive finding.
3. **Expansion to Include Segmentation:** As discussed, the current system performs classification. A powerful extension would be to incorporate a segmentation model (such as a U-Net) to precisely outline the area of pathology, which would allow for the quantitative measurement of disease severity.
4. **Implementation of an Asynchronous Task Queue:** To improve scalability for a high volume of users, the AI processing pipeline should be re-architected to use a proper asynchronous task queue (e.g., Celery with Redis). This would allow the API to return an immediate response while the AI analysis runs as a background job.
5. **Integration with Hospital Information Systems (HIS):** Future work should explore integrating the platform with existing hospital systems using standards like DICOM and HL7/FHIR to create a seamless clinical workflow.

5.5 Conclusion

This research successfully demonstrated the feasibility and effectiveness of a novel, intelligent multi-agent AI architecture for multi-disease medical image analysis. We designed and built a complete, end-to-end system that is not only diagnostically accurate but also interpretable, secure, and designed for continuous improvement. The project successfully addressed the identified research gap by creating a versatile, multi-agent platform that mimics a clinical escalation workflow to manage uncertainty. While further clinical validation is required, this work represents a significant step forward and provides a robust foundation for the future development of safe and reliable AI-powered tools for healthcare in Ghana and beyond.

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